ACADEMY OF TECHNOLOGY

Internal Examination, Even Semester, 2025

Paper name: Programming For Problem Solving | **Paper Code: ES-CS201**
Discipline: CSE /CSBS /ME /ECE /EEE /EE | **Semester: 2nd**
Time: 1 Hour | **Full Marks: 25**

Group A

Attempt any five questions from the following. **[1 × 5 = 5]**

1. i) Which of the following is ternary operator?
   a) + b) ++ c) ?: d) = =

   ii) What is the value of variable b?
   ```
   int b=printf("ES-CS201");
   ```
   a) 8 b) 6 c) 7 d) garbage value

   iii) Which will be the **address of** operator among the following?
   a) & b) * c) = = d) &&

   iv) What will be the predicted output of the following programming snippet?
   ```
   if (-24)
       printf("HELLO");
   else
       printf("WORLD");
   ```
   a) HELLO b) HELLOWORLD c) WORLD d) Compilation Error

   v) `printf("%d", -5 && printf("HELLO"));`.....gives
   a) 0 b) HELLO1 c) 1 d) 1HELLO

   vi) Post-test loop refers to:
   a) for loop b) while loop c) do-while loop d)Both of (a) and (b)

   vii) What is the output if we consider the following programming code?
   ```
   #include<stdio.h>
   int main ( )
   { int arr[4] = {1, 2, 3, 4};
   printf("%d", 2[arr]); }
   ```
   a) 1 b) 2 c) 3 d) 4

Group B

Attempt any four questions from the following **[4 x 5=20]**

2. Applying an iterative approach, write a C program to check whether the given number is Palindrome or not. **[5]**

3. Differentiate entry-controlled loop and exit controlled loop by showing control flows of such loops with a suitable example. **[5]**

P.T.O

4. Using loop control structure, how do you print (through C programming code) the following pattern up to a positive inputted integer (greater than 1)? [5]

```
    1
  1 3 5
1 3 5 7 9
```

5. State the impact of an array on the programming field. Is it necessary to give a break statement after every switch cases? Explain briefly with appropriate example. [2+3 = 5]

6. Compare if-else and switch in terms of C programming language. Predict and write the consequence if modulus operator % is used between two float variables. [3+2 = 5]

7. Demonstrate the procedure of linear search through a C programming code. [5]

ACADEMY OF TECHNOLOGY

Internal Examination, EVEN Semester, 2025

Paper Name: English | **Paper Code: HM-HU201**
Discipline: CSE/CSBS/ECE/EE/ME/EEE | **Semester: 2nd**
Time: 1 Hour | **Full Marks: 25**

Group -A

Attempt any five questions from the following: [1 x 5 =5]

i) What does the prefix "pre-" signify in the word "preview"?
a) After
b) Between
c) Before
d) Between

ii) Which words written on the tomb of a person:
a) epitaph
b) manuscript
c) inscription
d) Engrave

iii) Choose the word most similar in meaning to: ERADICATE
a) See
b) Happen
c) Remove
d) Accept

iv) Virat Kohli has penchant for hook shots. The synonym of 'PENCHANT' is:
a) Art
b) inclination
c) strength
d) desire

v) The word 'bite' and 'byte' are examples of:
a) Homonyms
b) blending
c) Homophones
d) Acronyms

vi) Arjun has a bachelor's degree from ____ University of Madras.
a) a
b) an
c) The
d) No article

vii) It is surprising indeed that he has an access ____ all the important leaders of the country.
a) About
b) Down
c) to
d) of

P..T.O.

Group -B

Attempt any four questions from the following: **[5 x 4 =20]**

2. Which are the instances when we use capital letter/s in written English?

3. Discuss e-mail etiquette.

4. Punctuate the following sentences:

a. The professor said please submit your assignments by Friday late submissions will not be accepted
b. I have three favorite genres of books mystery science fiction and historical fiction
c. My friends advice was simple focus on your goals stay determined and never give up
d. Although it was raining we decided to go for a walk which in hindsight was a bad idea
e. John couldn't believe his luck he had won the competition received a scholarship and was invited to a prestigious event

5. Rearrange the following sentences in the proper sequence to form a meaningful paragraph and then write the correct order.

a.One evidence of this is that many highly intelligent people are very low on creativity.
b.In other words, they may find replicating a process very easy but creating a process difficult.
c.But it is unfortunate that in many organizations these replicates are rated high and promoted over creators.
d.Intelligence and creativity are two different things.
e.They may understand the things very fast but are poor if any new thing is to be developed; this in the long-term gives creators an upper edge.

6. Discuss some techniques for achieving cohesion in writing.

Academy of Technology

Internal Examination, 2nd Semester, 2025

Sub: Chemistry-I **Discipline: CSE** **Code: BS-CH201**

Time: 60 min. **Full Marks: 25**

Use the following information if required:

$m_e = 9.11 \times 10^{-28}\ gm, \quad h = 6.625 \times 10^{-27}\ erg.s, \quad c = 3 \times 10^{10}\ cm/s.$

Group A (Attempt any <u>five</u> questions from the following) **[1 × 5 = 5]**

1. (i) An electron is confined in a 1D potential well of length 0.6 nm. The zero-point energy is
(a) 2.345 eV, (b) 1.045 eV, (c) 1.723 eV, (d) 2.164 eV.

(ii) Which one of the following is the frequency range for fingerprint region of IR spectroscopy?
(a) $40 - 100\ cm^{-1}$, (b) $500 - 1500\ cm^{-1}$, (c) $1000 - 2500\ cm^{-1}$, (d) $50 - 500\ cm^{-1}$.

(iii) For a given Eigen function $\psi = ae^{ikx}$, which one of the following represents the Eigen value of the operator $(-\nabla^2)$?
(a) aik, (b) k^2, (c) ik, (d) $-k^2$.

(iv) Which one of the following relations is correct?
(a) $\bar{\nu} = 2B(J+1)$, (b) $\bar{\nu} = 2\bar{B}(J+1)$, (c) $E_J = 2\bar{B}(J+1)$, (d) $E_J = 2BJ(J+1)$.

(v) For a given Eigen function ψ, which one of the following is true?
(a) The function ψ is not a normalized function.
(b) If $\frac{d\psi}{dx} = 0$, $\frac{d}{dx}$ is an Eigen operator.
(c) The function ψ cannot be differentiable within a given volume under consideration.
(d) In a given volume under consideration, there must be a point where ψ has two values.

(vi) Energy difference between 4th rotational level and 2nd rotational level is
(a) $12B$, (b) $14B$, (c) $2B$, (d) $6B$.

(vii) Which one of the following relations is correct?
(a) $2\pi\bar{\nu} = \sqrt{(K/\mu)}$, (b) $2\pi c\bar{\nu} = \sqrt{(K/\mu)}$, (c) $2\pi c\nu = \sqrt{(K/\mu)}$, (d) $\nu = \sqrt{(K/\mu)}$

Group B (Attempt any <u>four</u> questions from the following) **[5 × 4 = 20]**

2. State the primary requirement of a molecule to be microwave-active. Give two examples each of microwave-inactive and microwave-active molecules. (5)

3. "The frequency of vibration of a bond decreases with increase in masses of atoms" – Is it True or False? Justify your answer. "Stretching frequency of $C \equiv C$ less than that of $C = C$" – Is it True or False? Justify your answer. (5)

4. An electron, confined in a 1D potential well of length L, absorbs radiations of frequency ν and gets excited from $n = 1$ state to $n = 2$ state. If the box length is doubled, calculate the frequency (in terms of ν) needed to produce the same transition ($n = 1$ to $n = 2$). (5)

5. "Acetylene has no permanent dipole moment and hence it is IR-inactive" – Is it True or False? Justify your answer.
What type of molecule shows peaks in Fingerprint region? Give two examples. (5)

6. Write down the expression of energy for J^{th} rotational energy level. Hence, show that $\bar{\nu} = 2\bar{B}(J+1)$, where, the symbols have usual significances. (5)

7. Write down the expression of The Hamiltonian operator. "Zero-point energy of a tiny particle is not zero" – Is it True or False? Justify your answer using Heisenberg's uncertainty principle. (5)

ACADEMY OF TECHNOLOGY

Internal Examination, Even Semester 2025

Stream: CSE/CSBS **Semester: 2ND**

Time: 1 hr. **Marks: 25**

Subject: Mathematics IIA **Subject Code: BSM-201**

Group -A

Attempt any five questions from the following: **[5 x 1 =5]**

1. The probability that a leap year selected at random will contain 53 Sundays is
   a) 2/53 b) 52/53 c) 1/7 d) 2/7
2. The variance of a random variable X is
   a) $[E(X)]^2$ b) $E(X^2)$ c) $E(X^2) - [E(X)]^2$ d) $E(X^2) - E(X)$
3. For a random variable X, Var(X) = 1 then the value of Var(2X + 3) is
   a) 2 b) 4 c) 7 d) 5
4. The distribution for which mean and variance are equal to
   a) Poisson b) Normal c) Binomial d) Exponential
5. The mean of the Binomial distribution is np. Then its variance is
   a) npq b) npq^2 c) np d) np/q
6. If A and B are independent then A^c and B^c are also independent
   a) True b) False
7. If $A \subset B$ then P (B – A) is equal to
   a) P(A) - P(B) b) P(B) – P(A) c) P(A) – P(AB) d) P(B) –P(AB)

Group -B

Attempt any four questions from the following: **[4 x 5 =20]**

2. For two events A and B if P() = 5/8, P(AB) = 7/8, $P(A \cap B)=1/4$ then find the conditional probability of A on the hypothesis that B does not occur.

3. A bag contains 4 white and 2 black balls and a second bag contains 3 of each colors. A bag is chosen at random a ball is then taken out of it. What is the probability that the selected ball is a white?

4. In a bolt factory, the machines M1, M2 and M3 manufacture respectively 25, 35 and 40 percent of the total product respectively. Of their output 5, 4 and 2 percent respectively are defective bolts. One bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by machine M3?

5. A car-hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with average number of demand per day 1.5. Calculate the proportion of days on which neither car is used and the proportion of days on which some demand is refused. ($e^{-1.5} = 0.2231$)

6. A random variable X has the following probability function:

X	0	1	2	3	4	5	6	7
P(x)	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

Find the value of K and estimate P (X< 6) and P (0<X<5).

7. The mean and variance of binomial distribution is 3 and 3/2. Find the probability of obtaining atmost 3 successes.